ZETECH UNIVERSITY

ZET
Zetech Innovation,Entreprenuership & Technology Hub

Zetech University Research & Innovation Club

Discover. Innovate. Transform the Future

Club Portfolio 2026

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Foreword

Foreword by the Chairperson

On behalf of the Zetech University Research & Innovation Club, I am pleased to present the 2026 Research & Innovation Club Portfolio. This document reflects the collective effort, growth, and commitment of our members to advancing research, innovation, and impactful problem-solving within and beyond the university community.

The portfolio has been carefully developed and systematically structured to provide a comprehensive overview of the club's vision, leadership, resources, and achievements. It documents both flagship and emerging projects - **BIT, BLUE, CampusHub, ChatPay, Fourt, Mobi, Sentient, Umi, Sentry Hub** and **Wecket,** each representing our members' dedication to addressing real-world challenges through technology, research, and entrepreneurship. These projects align strongly with the United Nations Sustainable Development Goals and demonstrate the club's contribution to sustainable development, innovation, and community impact.

In addition to project documentation, this portfolio highlights the club's participation in competitions, ongoing and proposed research initiatives, and a comparative reflection on our progress over time. It also outlines upcoming activities, underscoring our commitment to continuous growth, collaboration, and excellence in innovation.

As Chairperson, I am immensely proud of the strides the club has made and the culture of curiosity, collaboration, and discipline that continues to define our work. This portfolio serves not only as a record of our achievements but also as a statement of intent—our resolve to nurture talent, foster interdisciplinary collaboration, and position Zetech University as a hub for impactful research and innovation.

I extend my sincere appreciation to the club leadership, members, mentors, and partners whose support and dedication have made this work possible. We look forward to continued collaboration and greater impact in the years ahead.

Moses Tabalya
Chairperson
Zetech University Research & Innovation Club
2026

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Executive Summary & Club Overview

The Zetech University Research & Innovation Club brings together students across faculties to research, innovate, and develop impactful solutions. Our project portfolio demonstrates strong alignment with global sustainability priorities, student capacity building, and community impact.

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Zetech University Research & Innovation Club

Club Leaders



Md. Esther Mwangi
Patron
Zetech University



Moses Tabalya
Chairperson
Zetech University



Elly Okello
Interim-Chairperson
Zetech University



Paul Wako
Vice Chairperson
Zetech University



Ivy Nganga
Programme Cordinator
Zetech University



Gabriel Nyaringo
Communication Director
Zetech University



Emmanuel Bett
Membership Chair
Zetech University

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Members

Names	Admission No.	Contact
1. Aaron Rop	1. **BSE-01-0082/2025**	1. **0748881693**
2. Abdi Aziz Edin	2. **BSE-01-0155/2025**	2. **0722795438**
3. Ahmed. A Sheikh	3. **BSE-01-0122/2025**	3. **0719525568**
4. Alex Kinyua	4. **BSE-01-8266/2023**	4. **0700531882**
5. Angel Veronica	5. **DEE-01-004/2025**	5. **0743042166**
6. Anne Wanjiru	6. **DEEZ-01-0022/2025**	6. **0113723758**
7. Aisha Isaa	7. **DHM-01-0017/2025**	7. **0768602446**
8. Benson Njoroge	8. **DEF-02-0004/2026**	8. **0702965968**
9. Beres Nganga	9. **CICTZ-01-4094/2022**	9. **0792492946**
10. Beth Waweru	10. —	10. **0726381472**
11. BILLY MBASILU	11. **DSE-01-8589/2024**	11. **0738830875**
12. Brandon Chomba	12. **BSE-01-0027/2026**	12. **0793670697**
13. Brian Juma	13. **BSE-05-0281/2023**	13. —
14. Caleb Carson Anjila	14. **BSE-05-0090/2025**	14. **0794545405**
15. Carlos Lichoti	15. —	15. **0732960072**
16. Cedric Onyango	16. **BMDC-05-0355/2023**	16. **0793403721**
17. Collins Munene	17. **DCF-03-0133/**	17. **0743976022**
18. David Soita	18. **BSCIT-01-0016/2023**	18. **0748264793**
19. David Wachira	19. **DSE-01-0500/2025**	19. **0741085552**
20. Dennis Njue	20. **BSE-05-0219/2025**	20. **0757345102**
21. Dwayne Njoroge	21. **DCF-01-0014/2026**	21. **0702708461**
22. Edwin Wanjau	22. **BCS-03-0334/2025**	22. **0748211804**
23. Elly Okello	23. —	23. **0769182575**
24. Emmanuel Bett	24. **DCS-01-8576/2024**	24. **0746895722**
25. Emmanuel Shikanda	25. **DCS-01-0271**	25. **0707215959**
26. Ephraim Kamau	26. **DSE-01-0434/2025**	26. **0717101762**
27. Eric K. Kangara	27. **DSE-01-0436/2025**	27. **0768554031**
28. Ethan Muthoni	28. **BSCIT-05-0057/2025**	28. **0117487475**
29. Felix Omondi	29. **BHTM-01-0013/2024**	29. **0717523999**
30. Gabriel Nyaringo	30. **BSE-05-0101/2025**	30. **0714010542**
31. George Boro	31. **DEE-01-0021/2025**	31. **0745301457**
32. Gift Muthusi	32. **CICTZ-01-0180/2025**	32. **0724804297**
33. Isaac Muendo	33. **BSE-01-0024/2025**	33. **0715552889**
34. Isaac Nugre	34. **DSE-01-8562/2024**	34. **0745676466**
35. Ismael Abdullahi	35. **BSE-01-0297/2023**	35. **0768441099**
36. Ivy Nganga	36. **DDA-01-0047/2025**	36. **0701230993**
37. Ivan Kubasu	37. **BSCIT-05-0180/2021**	37. **0114872974**
38. James Nkaayio	38. **DCS-01-8652/2024**	38. **0757060751**
39. Janet Kaluki	39. **DHM-01-0048/2024**	39. **0798388719**
40. Jeff Waithaka	40. **CICTZ-01-0115/2023**	40. **0746976707**
41. Jessy Odhiambo	41. **CEF-01-0019/2026**	41. **0704747593**
42. John Wen	42. **CICTZ-01-0171/2025**	42. **0711077619**
43. Joseph Nganga	43. **DEF-01-0005/2025**	43. **0799378037**
44. Julius Gwakole	44. —	44. **0708224990**
45. Keith Nyamai	45. **BMDC-01-0009/2024**	45. **0719238827**
46. Kirsten Maina	46. **DHM-02-0089/2024**	46. **0799650419**
47. Levis Kiprop	47. **BSE-05-0074/2024**	47. **0713440557**
48. Levis Njeru	48. **DSE-01-8360/2023**	48. **0797103037**
49. Levy Chege	49. **BAF-05-0106/2024**	49. **0791318491**
50. Linnet Muchemi	50. **BSE-01-0069/2026**	50. **0746685572**
51. Margues K. Gachanja	51. **CICTZ-01-0168/2024**	51. **0706143302**
52. Mark Kivuva	52. **DAC-01-0028/2024**	52. **0116318517**
53. Mary Mutheu	53. **DHM-01-0096/2024**	53. **0721159981**
54. Maureen Maina	54. **CICTZ-01-0174/2025**	54. **0741720313**
55. Mburu Karanja	55. **BSCIT-05-0113/2022**	55. **0797605689**
56. Mercy Limo	56. **DCS-03-0020/2026**	56. **0719499798**
57. Mercy Ochieng	57. **BSE-05-0143/2022**	57. **0700250542**
58. Michael Oginda	58. **DSE-01-0237/2025**	58. **0740912144**
59. Miriam Wakesho	59. **CPAZ-01-0003/2026**	59. **0790717657**
60. Moses Tabalya	60. **DIT-01-0044/2025**	60. **0719806803**
61. Naomi Kerubo	61. **DPSM-02-0083/2025**	61. **0759450290**
62. Nyaigotti Nelson	62. **BSE-01-0053/2024**	62. **0768560186**
63. Orenge James Junior	63. **BSE-05-0049/2025**	63. **0746415801**
64. Patrick Wateri	64. **BSE-05-0176/2022**	64. **0715835657**
65. Peter Mwanita	65. **DDA-02-0109/2025**	65. **0727003182**
66. Purity Njeri	66. **BHTM-01-0007/2026**	66. **0724942540**
67. Reuben Kamau	67. **DSE-01-0105/2025**	67. **0799031535**
68. Rioba Alex	68. **BSCIT-05-0037/2019**	68. **0707626068**
69. Rodgers Owino	69. **DDA-01-0076/2025**	69. **0710908348**
70. Romeo Mugaya	70. **DSE-01-0397/2025**	70. **0717071264**
71. Ryan Onyango	71. **DDM-02-0064/2025**	71. **0782398507**
72. Samuel Mwae	72. —	72. **0743213926**
73. Samuel Mwandami	73. **CEE-01-0018/2025**	73. **0114770994**
74. SAMUEL ODUOR	74. **BSCIT-05-0353/2022**	74. **0701722066**
75. Sammy Njenga	75. **DBIT-01-0092/2025**	75. **0771011656**
76. Selovas Losen	76. **DHM-01-0107/2024**	76. **0748165886**
77. Shannies Wahome	77. **BBAM-01-0012/2026**	77. **0718813867**
78. Sharon Mwende	78. **DHM-01-0057/2024**	78. **0796326052**
79. Sophia Michelle	79. —	79. **0701805643**
80. Stacy Wandia	80. **DHM-01-0049/2025**	80. **0792209115**
81. Tasha Liavuli	81. **DCF-02-0028/2024**	81. **0113479388**
82. Tecla Salem	82. **DSE-01-8580/2024**	82. **0798767217**
83. Timothy Minani	83. **DIT-01-8150/2023**	83. **0769295800**
84. Tony Gitonga	84. **DIT-01-0003/2025**	84. **0112300049**
85. Tynah Wambui	85. **DCS-01-0006/2025**	85. **0711271275**
86. Victoria Theuri	86. **BBIT-05-0109/2022**	86. **0799142010**
87. Victor Mwrigi	87. **DSE-01-0386/2025**	87. **0757267713**
88. Zakbravo Wanyekia	88. **CEEZ-0-0061/2024**	88. **0741088373**
89. Felix Omondi	89. **BHTM-01-0013/2024**	89. **0717523999**
90. Gabriel Nyaringo	90. **BSE-05-0101/2025**	90. **0714010542**
91. Brandon Chomba	91. **BSE-01-0027/2026**	91. **0793670697**
92. Benson Njoroge	92. **DEF-02-0004/2026**	92. **0702965968**
93. Linnet Muchemi	93. **BSE-01-0069/2026**	93. **0746685572**
94. Mercy Limo	94. **DCS-03-0020/2026**	94. **0719499798**
95. Purity Njeri	95. **BHTM-01-0007/2026**	95. **0724942540**
96. Shannies Wahome	96. **BBAM-01-0012/2026**	96. **0718813867**
97. Dwayne Njoroge	97. **DCF-01-0014/2026**	97. **0702708461**
98. Miriam Wakesho	98. **CPAZ-01-0003/2026**	98. **0790717657**
99. Jessy Odhiambo	99. **CEF-01-0019/2026**	99. **070474759**

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Volunteering Members

ACTIVITIES 2026
ROBOTICS FOR GOOD
YOUTH CHALLENGE

Names	Admission No.	Contacts
1. Collins Munene	1. DCF-03-0133/	1. 0743976022
2. Elly Okello	2. —	2. 0769182575
3. Ephraim Kamau	3. DSE-01-0434/2025	3. 0717101762
4. Ismael Abdullahi	4. BSE-01-0297/2023	4. 0768441099
5. Larry Kayeli	5. DSE-01-0565/2023	5. 0743804958
6. Levis Njeru	6. DSE-01-8360/2023	6. 0797103037
7. Naomi Kerubo	7. DPSM-02-0083/2025	7. 0759450290
8. Nyaigotti Nelson	8. BSE-01-0053/2024	8. 0768560186
9. Orenge James Junior	9. BSE-05-0049/2025	9. 0746415801
10. Reuben Kamau	10. DSE-01-0105/2025	10. 0799031535

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Mission & Vision

Mission: Explore, research, and push boundaries to nurture innovation in a supportive environment.
Vision: Collaborate, learn, and create impactful projects to unlock individual potential and shape the future.

Membership Overview

The club has experienced significant growth, with over 200 members recruited, reflecting increased student interest and engagement in innovation.

Club Resources

The club provides access to funding opportunities, mentorship programmes, apprenticeship platforms, and financial management support, enabling members to transition ideas into viable solutions.

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All club-based projects are intentionally designed to align with the United Nations Sustainable Development Goals (SDGs), ensuring that each initiative addresses corporate social responsibility (CSR) priorities and delivers measurable community impact. This strategic approach enables the club to maximize social, environmental, and economic outcomes while efficiently utilizing limited resources.

1. BIT

BIT (Biometric Management System)

Background & Context:
BIT was created to solve challenges in identity verification for institutions, events and personal and institutional performance monitoring in Kenya. BIT'S diverse team is also involved in product design and develoment ranging to smart infrastructure and IoT devices to fit into and lead Kenya's Technological development and Innovation Manual identification systems are prone to fraud, human error, and inefficiency. BIT leverages AI-powered systems facial recognition and fingerprint scanning to provide fast, accurate, and secure authentication, making it ideal for universities, corporate offices, and public gatherings. Since its inception in June, 2025, BIT has been a student-led innovation emphasizing safety, efficiency, and technological advancement.

Technical Architecture & Features:
BIT integrates multi-modal biometrics: facial recognition using Python-based AI algorithms, fingerprint scanning, and QR-code verification. Data is encrypted with AES-256 standards, ensuring privacy and security. A cloud-based backend allows scalable storage, while a real-time monitoring dashboard provides automated alerts for unauthorized access. Additional features include attendance tracking, analytics dashboards, and integration with existing HR and event management systems.

Operational Model & Market Fit:
BIT follows a B2B SaaS deployment model, offering subscription tiers and enterprise licenses. Universities, government offices, and event organizers benefit from modular deployment, where smaller organizations can use core features, and larger ones can adopt enterprise analytics and multi-site integration.

Societal & SDG Impact:
BIT contributes to SDG 11 (Sustainable Cities and Communities) by enhancing public safety and SDG 8 (Decent Work and Economic Growth) by creating over 150 direct jobs in technical support and AI training, plus 500+ indirect jobs in implementation and partnerships.

Projected Impact & Future Growth:
Projected first-year revenue is KES 9,750,000. Expansion plans include East Africa integration, mobile-based verification, predictive AI analytics for security, and NFC-enabled ID solutions.

Team:
Moses Tabalya
Elly Okello
Joseph Okeyo
Beth Waweru
Emmanuel Bett
Ruth Munyiri - TUK
Novall Smollo
Biran Juma

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Illustrations



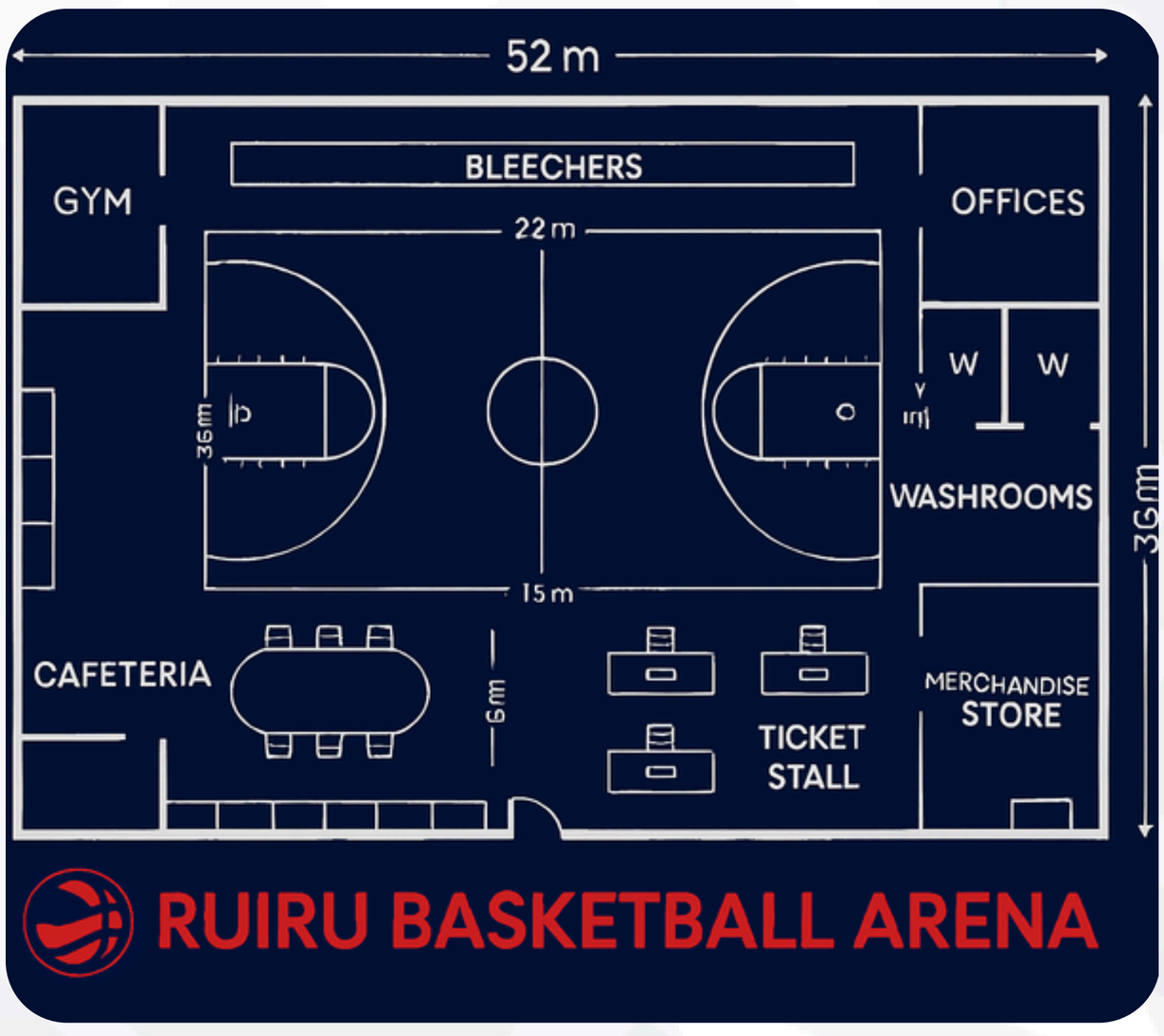
Surveillance drone concept



Athlete performance tracking device

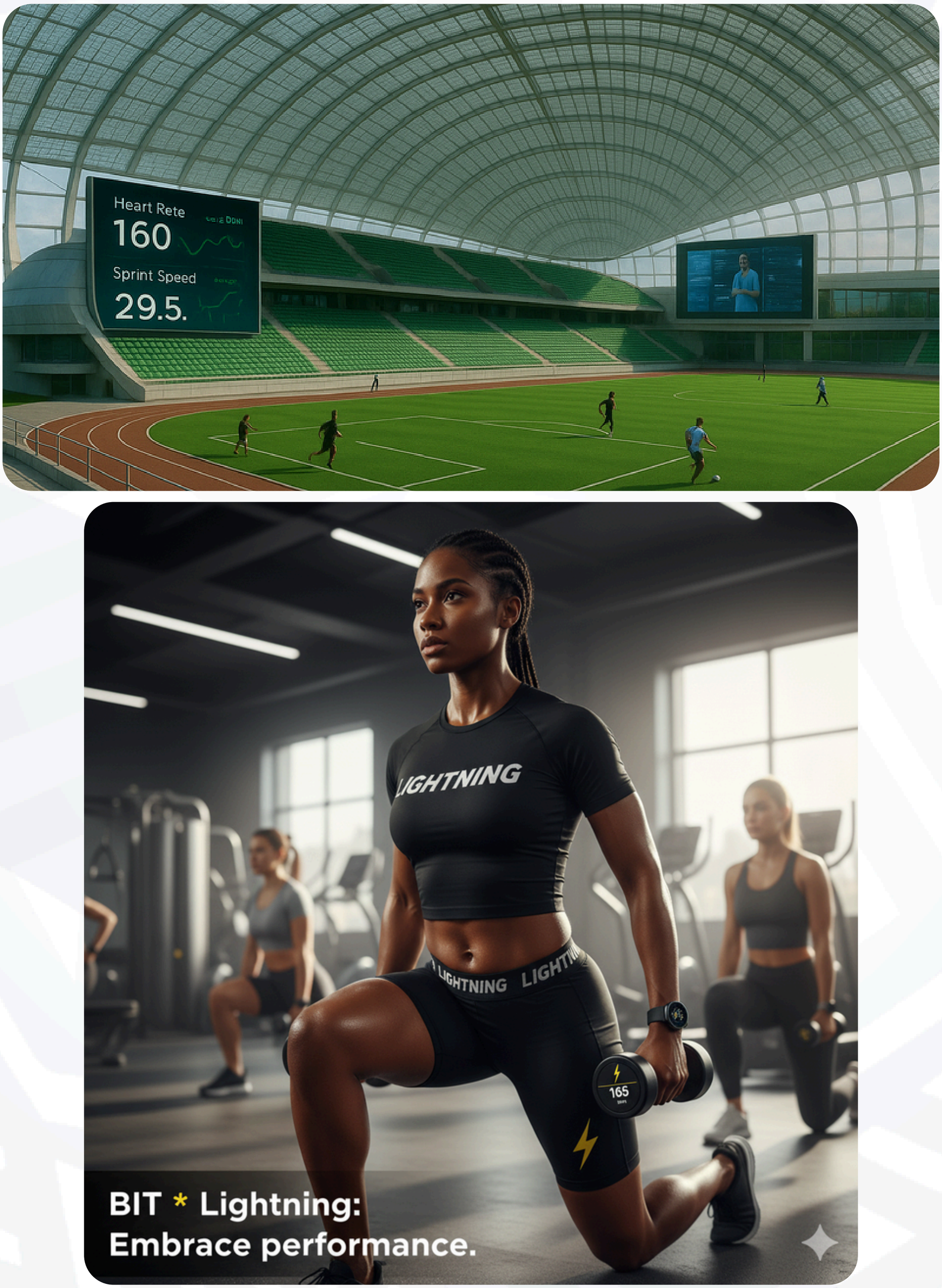


AI/ML Powered Cameras



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Illustrations



Conceptualized BIT’s devices in use

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Use case:

All of BIT’s products and services revolve around each other in a single diverse ecosystem giving our users a complete package that boosts their productivity, safety, and convinience, at work and at home seemlessly.

2. BLUE (Ocean Intelligence Platform)

Background & Context:
BLUE aims to address environmental challenges affecting oceans and fisheries. Kenya's coastal communities face declining fish stocks, pollution, and inefficient maritime monitoring. BLUE collects real-time oceanographic, weather, and marine traffic data to inform decisions for sustainability and conservation.

Technical Architecture & Features:
The platform combines IoT sensors, satellite imagery, and AI analytics to monitor water quality, biodiversity, and maritime activity. Users can access dashboards with predictive models, alerts for illegal fishing, and recommendations for sustainable practices.

Operational Model & Market Fit:
BLUE targets governmental agencies, environmental NGOs, maritime companies, and coastal communities. Revenue is generated through data subscriptions, consultancy, and partnerships with fisheries.

Societal & SDG Impact:
BLUE aligns with SDG 14 (Life Below Water) and SDG 13 (Climate Action), reducing overfishing, promoting sustainable practices, and preserving marine biodiversity. It also generates high-value analytical jobs in environmental monitoring.

Projected Impact & Future Growth:
Expected to improve marine conservation efforts, support sustainable fisheries, and expand to regional African waters within five years, creating both local employment and international collaboration opportunities.

Team:
Moses Tabalya
David Soita

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3. CampusHub

CampusHub streamlines student life management in Kenyan universities, addressing the fragmentation of services like accommodation, event updates, and peer collaboration.

Technical Architecture & Features:
CampusHub integrates student portals, real-time event tracking, hostel bookings, and social networking within a mobile-friendly interface. Features include notification systems, event calendars, and interactive forums for collaboration.

Operational Model & Market Fit:
Primarily for university students and staff, it generates revenue via subscription services for premium features, partnerships with campus service providers, and institutional licensing.

Societal & SDG Impact:
Supports SDG 4 (Quality Education) and SDG 11 (Sustainable Cities and Communities) by enhancing campus living experiences, improving access to educational resources, and fostering collaboration.

Projected Impact & Future Growth:
Expected to onboard over 10,000 active student users within two years, improve campus service efficiency by 40%, and scale to multiple universities across Kenya.
Team:
Elizer Mwachugha & James Junior

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4. ChatPay

Background & Context:
ChatPay is a social payments app that allows users to personalize financial transactions on social platforms. This project addresses the gap in peer-to-peer social commerce and micro-payments.

Technical Architecture & Features:
The app integrates secure payment gateways with social media APIs, enabling seamless money transfers, gifting, and micro-transactions within chat interfaces. It also includes AI-based transaction recommendations and analytics dashboards for users.

Operational Model & Market Fit:
Targeting social media users, micro-influencers, and e-commerce communities. Monetization includes transaction fees, premium features, and potential partnerships with payment providers.
Societal & SDG Impact:
Contributes to SDG 8 (Decent Work and Economic Growth) by enabling micro-entrepreneurship and digital financial literacy. Also promotes SDG 9 (Industry, Innovation, and Infrastructure) by fostering innovative fintech solutions.
Projected Impact & Future Growth:
Patent process underway; expected to reach 50,000 users within the first year, with potential integration across multiple platforms to enhance peer-to-peer commerce.
Project Lead: Paul Wako

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5. Mobi

Mobi – Mobile Transportation Platform Portfolio Overview
Project Overview
Mobi is a comprehensive urban mobility platform designed to unify Kenya's transportation ecosystem. By integrating public transport ("matatus"), private ride-hailing services, campus shuttles, and eco-friendly alternatives, Mobi offers a seamless, user-friendly solution for students, professionals, and daily commuters. The platform addresses inefficiencies in reliability, fragmented services, and lack of real-time information.

Vision
To be the most trusted and comprehensive mobile transport platform in Kenya, empowering every individual with simple, reliable, and safe travel solutions.

Mission
To connect people with their destinations through a seamless, all-in-one digital platform that simplifies travel, reduces wait times, and enhances security for every journey.

Project Development Status (As of 2025)

- Project concept finalized
- Business plan drafted
- Frontend mobile/web development initiated
- Backend, real-time tracking systems, and API integrations pending
- Matatu operator onboarding and regulatory approvals pending
- Pilot deployments and beta testing pending

Project Team & Roles

- Moses Tabalya – Project Lead & Technical Architect
- Joseph Okeyo – App Development & Frontend Lead
- Beth Waweru – Marketing & Communications Lead
- Eve Wangui – Operations & Customer Support
- Billy Basilu – Security & Compliance Coordinator
- Emmanuel Bett – Backend Systems Developer
- Rosemary Kariuki – Business Development & Partnerships

Market Analysis

- Industry Overview: Kenyan road transport dominates GDP contribution, with matatus and ride-hailing as primary modes. Fragmented services leave opportunities for consolidation and digital innovation.
- Target Market:
  - Students: Campus shuttles and group rides
  - Daily Commuters & Professionals: Public transport and premium ride services
- Competitive Advantage:
  - Single platform integrating public, private, and student transport
  - Unique matatu sector integration
  - User-centric features tailored to local transport pain points

- Revenue Streams:
  - Transaction fees per ride
  - Subscription plans for frequent users/businesses
  - In-app advertising & partnerships
- Financial Projections: Detailed financial modeling to project revenue and profitability over 3–5 years (sep document)

6. RevvWash

Background & Context
RevvWash addresses the challenge of limited access, long queues, and time inefficiencies associated with traditional car wash services. With increasing urban mobility demands and busy lifestyles, there is a growing need for flexible, location-independent vehicle care solutions that bring services directly to the customer.
Technical Architecture & Features
The system is built as a mobile-first digital platform supported by GPS-enabled mobile service units. It integrates booking, geolocation, distance-based pricing, and digital payments, enabling real-time service matching between customers and mobile car wash vans.
Operational Model & Market Fit
RevvWash operates using a mobile service model where equipped vans are dispatched to customer locations on demand. This model fits well in urban environments like Nairobi, where convenience, time savings, and flexible service access are highly valued by vehicle owners.
Social Impact & Alignment with UN SDGs
The project supports sustainable development by creating employment opportunities, promoting innovation in service delivery, optimizing resource usage, and contributing to smarter urban mobility aligning with the following Social Development Goals;
8-Decent Work and Economic Growth
9-Industry,Innovation and Infrastructure
11-Sustainable cities and Communities
12-Responsible consumption and Production

Project Impact & Future Growth
RevvWash improves accessibility to car care services while reducing time and travel costs for users. Future growth includes regional expansion, eco-friendly wash technologies, data-driven optimization, and integration with broader smart mobility services.

Project Lead: **Nyaigotti Nelson**

7. Sentient

Sentient Project – Portfolio Overview
Project Overview
Sentient (Geo-Resilience AI Platform) is a predictive urban intelligence system designed to transform Nairobi into Africa's most resilient city. By integrating NASA space data, AI, and machine learning, Sentient creates a digital twin of Nairobi that forecasts crises, informs urban planning, and empowers citizens and policymakers to act proactively.

Core Mission
To leverage geo-spatial intelligence and predictive analytics for urban resilience, minimizing disaster risk, optimizing human health outcomes, and establishing Kenya as a leader in AI-enabled urban planning and space-ready technology.

Key Achievements
Phase 1: Concept & Planning (2025)
Idea Development – Digital twin concept, AI & NASA data integration model
Business Plan Drafted – Operational roadmap, technical requirements, and funding strategy
Team Formation Initiated – Core roles and responsibilities identified
Early App Prototypes – Wireframes for citizen-facing and planning dashboards
Phase 2: Development (Ongoing)
Data Pipeline Setup – NASA & local urban datasets integration pending
AI/ML Engine Design – Predictive modeling framework in design stage
App Features – Clean Coin Hub, Land Safety Check, Health Hotspots, Agri-Alerts
Stakeholder Outreach – Early government, urban planning, and civic partners identified

Innovation Highlights
Technical Innovation
1. Digital Twin of Nairobi – Full simulation of urban infrastructure and environmental conditions
2. AI-Powered Predictive Modeling – Forecasts disaster location, timing, and severity
3. NASA Data Integration – Real-time climate, land movement, and geographical risk inputs
4. Geo-Resilience App – Human-centered interface for citizens, planners, and builders

Social Innovation
1. Clean Coin Hub – Incentivizes recycling with tokens or airtime, improving waste collection efficiency
2. Land Safety Check – Ensures safe housing and infrastructure development
3. Health Hotspots & Agri-Alerts – Alerts communities to disease risk and precision agriculture opportunities
4. Job Creation & Skill Building – AI, space tech, and urban planning skill development

Project Milestones
Milestone 1: Concept Validation
- Duration: Q3 2025
- Deliverables: Business plan, technical blueprint, early wireframes
- Success Metrics: Concept approved, roadmap for AI & NASA integration drafted

Milestone 2: Data & AI Engine Design 🚧
- Target: Q4 2025 – Q1 2026
- Deliverables: Framework for predictive modeling, NASA data pipelines designed
- Success Metrics: Prototype algorithms for disaster prediction tested on historical data

Team:
Moses Tabalya
Abubakar Kitheka
Armstrong Opondo

8. Umi

Project Overview
Umi is a student-focused digital planner and IP management platform designed to help students document, protect, and manage their ideas while streamlining academic and daily planning. The platform integrates FastAPI, Svelte, MySQL, and ML/AI-powered features for document analysis, version control, and intelligent workflows.

Core Mission
To provide a secure, intuitive, and efficient platform for student innovators, combining creation-to-protection workflow tools with smart scheduling, collaboration, and actionable insights.

Team & Roles
Moses Tabalya
Abubakar Kitheka
Armstrong Opondo

9. Wecket

Project Overview
Wecket is a next-generation social media platform designed to foster authentic community engagement through a decentralized, interest-based content model. Integrating Nebular AI, the platform provides personalized content curation, real-time chat, and a premium ad-free experience with the Golden Ticket subscription. Wecket prioritizes meaningful interaction, privacy, and positive digital experiences.

Core Mission
To empower individuals and communities to connect over shared interests using intelligent technology, creating a secure, personalized, and vibrant digital space.

Key Achievements
Phase 1: Foundation (2025)
✅ Concept Development – Platform idea and AI integration model finalized
✅ Business Plan Drafted – Full roadmap for operations, monetization, and scaling
✅ Frontend Development Initiated – HTML/XHTML/CSS prototypes underway
✅ Nebular AI Prototype – Early-stage AI content curation engine
Phase 2: Development (Ongoing)
🔄 Backend Development – Python + Firebase backend under construction
🔄 Community Onboarding Strategy – Initial beta planning
🔄 Premium Golden Ticket Model – Subscription & monetization features in progress
🔄 Regulatory Compliance & Data Security – Privacy and user protection protocols pending

Innovation Highlights
Technical Innovation
1. Nebular AI Engine – Personalized content recommendations and sentiment analysis
2. Decentralized Community Model – Interest-based feeds instead of algorithm-driven mass broadcasting
3. Premium Features – Ad-free experience, analytics, and customizable feeds via Golden Ticket subscription

Social Innovation
1. Community-Focused Design – Promotes authentic interaction over superficial engagement
2. User Privacy & Security – Transparent handling of data
3. Digital Literacy & Wellbeing – Encourages intentional, positive participation online

Project Milestones
Milestone 1: Platform Concept & Planning ✅
- Duration: Q3 2025
- Deliverables: Business plan, platform architecture
- Success Metrics: Concept validated, foundational roadmap completed

Milestone 2: Frontend Development Initiation 🚧
- Target: Q4 2025
- Deliverables: HTML/XHTML/CSS web prototypes
- Success Metrics: Wireframes completed, basic interactivity demonstrated

Milestone 3: Backend & AI Engine Development 📅
- Target: Q1–Q2 2026
- Deliverables: Python + Firebase backend, Nebular AI engine
- Success Metrics: Functional prototype capable of content curation

Milestone 4: Beta Community Launch 📅
- Target: Q3 2026
- Deliverables: Closed beta with early users, feedback loop integration
- Success Metrics: 500+ active beta users, Golden Ticket features tested

Team:
Moses Tabalya.
Washington Mwangi
Elly Okello
Joseph Okeyo
Team from BIT

10. Fourt

Project Portfolio Overview
Executive Summary Fourt is an innovation-driven technology company that incubates, develops, and delivers multiple solutions across diverse industries. As the mother company, Fourt provides the structure, resources, and technical expertise required to transform member-generated ideas into scalable, real-world services. Each idea becomes a Fourt Service Line, forming a connected portfolio spanning software systems, hardware innovation, civic technology, accessibility, and environmental sustainability. Mission To harness the creativity of young innovators to deliver practical, technology-based solutions that improve life, business, and governance. Vision To be Africa's most impactful innovation powerhouse — creating multiple services under one unified brand, Fourt.

Organizational Structure Mother Company: Fourt Limited – Provides funding access, mentorship, shared infrastructure, branding, and legal protection for all service lines.

- Fourt WasteTech: Smart garbage collection reminder system: Ruben Matoke.
- Fourt AgriLink: Digital marketplace for farmers, buyers, and transport: Ezra Kipkorir.
- Fourt SecureID: Biometric and smartcard authentication systems: Simon Walaka & Jamima Ngendo.
- Fourt SmartHomes: Secure smart doors and windows: Gitau Justin.
- Fourt Access: Assistive devices for visually impaired and deaf users: Novaal Smollo
- Fourt AquaClean: Technology for cleaning and preserving water bodies: Moses Tabalya
- Fourt Games: Video game development for entertainment and education: Washington Mwangi
- Fourt MobileGuard: Anti-phone theft technology: Billy Basilu.
- Fourt CivicTrack: Government expenditure tracking system: Maxwell Kibet.
- Fourt SmartRetail: Supermarket shopping assistance systems: Joseph Okeyo.

Impact:
Biometric and smartcard authentication systems Secure smart doors and windows Simon Walaka & Jamima Ngendo Gitau Justin Assistive devices for visually impaired and deaf users Novaal Smollo Technology for cleaning and preserving water bodiesMoses Tabalya Video game development for entertainment and education Washington Mwangi Anti-phone theft technology Government expenditure tracking system Supermarket shopping assistance systems Billy Basilu Maxwell Kibet Joseph Okeyo Fourt drives social, environmental, and economic impact by improving waste management, supporting agricultural value chains, enhancing accessibility for persons with disabilities, promoting environmental sustainability, strengthening security, and increasing transparency in governance.

Team:
Moses Tabalya
Novaal Smollo
Reuben Matoke, etc.

11. SentryHub

SENTRY Hub is a protective digital platform designed to safeguard vulnerable members of society from cyber crimes with a strong focus on scam prevention. The project provides awareness, reporting and early-warning mechanisms to help users identify, avoid and respond to scams occurring both online and offline including phone-based and face-to-face interactions. By combining education, community reporting and alert systems, SENTRY Hub empowers users to stay informed, act quickly and reduce the risk of financial, emotional and digital exploitation.

Team leader: Khadija Harun.

12. MinerAI

We Don't Just Analyze Maps.
We Build the Intelligence Layer for Earth.
MinerAI is a full-cycle geospatial AI development and deployment ecosystem. We transform raw satellite data into deployable AI systems, structured training datasets, and scalable spatial intelligence.
Not a GIS tool. Not a labeling environment. An infrastructure platform for building, deploying, and scaling earth intelligence.

Earth Intelligence Begins Here
minerAI is an end-to-end earth intelligence platform - extracting satellite data, modeling real-world systems, and delivering clear, defensible insights for industries reshaping the planet.
Team lead: Kelvin Murithi

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13. The Thinking Game

Abubakar Kitheka:

INTRODUCTION:
A Patent Management System is a software solution designed to streamline the management of patents, from submission to approval and tracking. With the increasing need for innovation and intellectual property protection, organizations require efficient systems to manage patent information securely and systematically. This project report presents the design, implementation, and evaluation of a comprehensive system aimed at simplifying patent management processes, ensuring data integrity, and supporting decision-making. The system integrates modern technologies to provide a scalable, user-friendly, and secure platform, demonstrating practical application of software engineering principles in solving real-world problems.

THE WHY BEHIND THIS PROJECT?
In a realm where ideas shape the future, the act of creation is both fragile and profound. Innovators and students alike are full of brilliant ideas ,moments of insight that could transform the way we think, learn, and build. Yet, these ideas often remain vulnerable: creators lack a formal system to document, track, and protect their intellectual property. The absence of a structured patent management process leads to difficulties in ownership verification, delayed approvals, and the risk of unintentional idea loss or misuse. To address this challenge, I built a digital Patent Management System that provides a secure, efficient, and transparent platform for recording, tracking, and managing innovations. The system enables ideas to be formally protected, reduces administrative inefficiencies, promotes innovation, and ensures that intellectual property rights are respected in academic and professional environments.

TOOLS OF TRADE
To bring the concept to life, a carefully chosen set of technologies was employed. The backend was built using Elixir and Phoenix, providing scalability, fault tolerance, and real-time capabilities. PostgreSQL was selected for the database, offering reliability and support for complex queries. The frontend leverages HTML, Tailwind CSS, and LiveView, 1 ensuring a responsive, intuitive interface. Each technology was chosen not only for its robustness but also for its alignment with the goals of clarity, efficiency, and maintainability.

CONCLUSION
The Patent Management System successfully addresses the challenges of recording, tracking, and protecting innovations. By providing a secure, transparent, and efficient platform, the system ensures that ideas are safeguarded, administrative processes are streamlined, and collaboration is made easy for innovators and reviewers alike. The modular design, real-time updates, and integration with modern tools demonstrate practical application of software engineering principles while offering scalability and maintainability for future growth. Beyond this project, I am actively pursuing initiatives to empower young tech 3 enthusiasts. I am currently building a Computer Science Club, Lock, where I try to modernise education in the age of AI . I am also working on developing my own innovative game, experimenting with AI-driven mechanics and creative design to create engaging experiences. These initiatives reflect my commitment to learning, innovation, and creating impactful solutions that inspire others. For further questions, feedback, or collaboration, I can be reached at: � ? Email: abubakar@craftinc.dev

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Activities: 2026

- **12TH JANUARY ANNUAL SUSTAINABLE BUSINESS COMPETITIVENESS SUMMIT 2026 - 2 Day (27th to 28th) - Ole Sereni Hotel, Nairobi.**
- **30TH JANUARY ENTREPRENEURSHIP EDUCATION SUMMIT - SINGLE DAY (30th) - Pullman Hotel, Upperhill, Nairobi.**
- **28TH FEBRUARY ONBOARDING CLUB MEMBERS TO THE WEZESHA INCUBATION PROGRAMME**
- **13TH MARCH; IDEATION SESSION & DESIGN THINKING WORKSHOP**
- **21ST MARCH ROBOTICS FOR GOOD YOUTH CHALLENGE**









2TH ANNUAL SUSTAINABLE BUSINESS COMPETITIVENESS SUMMIT 2026 - 2 Day (27th to 28th) - Ole Sereni Hotel, Nairobi.

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Activities: 2026









ENTREPRENEURSHIP EDUCATION SUMMIT - SINGLE DAY (30th) - Pullman Hotel, Upperhill, Nairobi.

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Activities: 2026

12th ANNUAL SUSTAINABLE BUSINESS COMPETITIVENESS SUMMIT 2026

The Research and Innovation Club participated in an SDG-aligned forum hosted by I Choose Life Africa, exploring inclusive and climate-resilient development in Kenya and Africa. Drawing from sustainability, entrepreneurship, and systems dialogue, the event highlighted clean energy, resilient infrastructure, circular economy solutions, and dignity-centered livelihoods. Emphasis was placed on innovation, community asset ownership, and partnerships that translate research into scalable impact.

Members:

1. Patron; Md. Esther Mwangi.
2. Chairperson; Moses Tabalya.
3. Member; Lewis Mwangi.
4. Member; Brian Mtana.
5. Member; Ismael Abdullahi.
6. Member; Kelvin Murithi

ENTREPRENEURSHIP EDUCATION SUMMIT - SINGLE DAY (30th) - Pullman Hotel, Upperhill, Nairobi.

The Entrepreneurship Education Summit by KENIA clearly highlighted the importance of entrepreneurship in the Kenyan economy and its alignment with Academia's and the Government's Initiatives to foster growth and innovation within the country and outside the region, it is coherent that there is collaboration to foster healthy Entrepreneurial growth and Innovation by developing sustainable ventures, this path will ensure Kenya stays in the path of Entrepreneurial development, technological advancement, and improved Educational Quality in the Nation enhancing Economic Growth in the region and beyond. The Key Guest clearly echoed this importance with a firm message and reminder from the President.

- Patron; Md. Esther Mwangi.
- Chairperson; Moses Tabalya.
- Member; Lewis Mwangi.
- Member; Brian Mtana.
- Member; Ismael Abdullahi.
- Member; Kelvin Murithi

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Activities: 2025

January 2025 · 20th Jan: Entrepreneurship Academy (Online Registration) · 29th Jan: Inclusive Africa Conference (Registration Deadline)

February 2025 · 6th Feb: GDG Community Dev · 16th Feb: Tech Divas · 17th Feb: Justice Entrepreneurship School Registration (Virtual) · 21st Feb: SME Innovation Challenge · 23rd Feb: International Youth Empowerment Foundation Cohort (Virtual) · 27th Feb: Ladies Empowerment (Auditorium, Ruiru Campus)

March 2025 · 1st March: Launchpad by Founders Connect · 7th - 8th March: Hack and Play (Mangu Campus) · 11th March: Men in Tech (TRC) · 12th March: Android Development Round Table (L204) · 21st March: Campus Hangout (Ruiru Campus) · 24th March: Collaboration with the Book Club · 26th March: The History and Future of AI Masterclass (Virtual) · 26th March: iSSTECH Info Session (Zetech University Mangu Campus)

April 2025 · 22nd April: Kenya Agribusiness Entrepreneurship Program · 29th April: 3rd Nairobi Startup Summit (Sarit Expo Centre) May 2025 ·

16th May: Ideas Festival (Zetech University Mangu Campus) · 19th May: Financial Planning and Management for Aspiring Entrepreneurs (Virtual) June 2025 ·

2nd June: Creator Competition (Online) · 6th - 7th June: Presidential Innovation Challenge and Award Training (Windsor Hotel, Nairobi) · 19th June: Financial Planning and Management Training (Online) · 24th June: Moringa and Walumu Hackathon · 26th June: Inter-University Hackathon, JKUAT (Online Registration) · 27th June: Cohort of ANZA Grow Academy July 2025 ·

1st - 31st July: Clean Tech Innovation Competition · 2nd July: 2025 DTB-KBA Financial Literacy and Enterprise Development Training for SMEs · 10th July: Campus Hangout (Ruiru Campus) · 11th July: AI and Emerging Technologies Workshop (Auditorium, TRC Thika Road Campus) · 15th - 16th July: World Youth Skills Day (KICC) · 18th - 19th July: Design Thinking Hackathon (Riara University) · 22nd - 23rd July: ANZA GROW (UNICAF Kenya Offices, West Park Suites) · 25th July: Innovation Day · 29th July: WIPO Young Experts Programme · July (Date TBA): ZFS Startup Grind, ZOHO (Online)

August 2025 · 1st August: British Council Entrepreneurship Council · 8th August: GitHub Workshop & Speaker Session (Zetech University, Mangu) · 12th August: Ye! Summit Youth Day (Virtual) · 25th August: Women in Tech September 2025 ·

12th September: Career Day · 15th September: Huawei Competition (Online) · 21st September:
Green Tech Hackathon (Online) October 2025 · 1st October: Project Creation and Management (Online) ·

2nd October: Microsoft for Start-up Resources (Online) · 5th - 6th October: NASA Space Apps Challenge (KALRO) · 8th October: Digital Internet Resilience and Safety (Hybrid: Online, Kibera Town Centre, Kamkunji, Kibera) · 9th October: Perplexity Pro Student Subscriptions (Online) · 9th October: Research Debate Themes Ideation (Online) · 10th October: ZUSA Awards · 13th October: How to Start a Startup (Online) · 15th October: Self-Paced Cisco Courses (Online) · 16th October: Career Fair · 16th - 17th October: AfriNovation Festival (ASK Dome, Nairobi) · 18th October: HackAGreenFuture Hackathon · 19th - 24th October: Zetech University Access 2.0 Student Innovation Challenge (The Marvel Hotel, Embu) · 22nd October: Nairobi Hacking Station (Zetech Mangu Campus) · 30th October: Ladies Forum (Auditorium) November 2025 ·

3rd Nov: Growth Africa Funding Opportunities (Online) · 9th - 10th Nov: WordCamp Nairobi 2025 (Daystar University) · 11th Nov: Uzuzi Hub + iZET Hub + ZURI Mentorship and Introduction Meeting (iZET HUB) · 11th Nov: Creation of Resources Pool (Online) · 13th Nov: Student Exchange Programme: Konza City · 13th - 14th Nov: ASSEK Annual Conference · 21st Nov: I Choose Life Podcast Meeting (ICL Offices)

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